Wine Transaction Anomaly Detection

Milestone 2

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Problem definition and implemented system

This system flags unusual wine purchase transactions for human review. A supervised wine cultivar model supplies a laboratory-to-declaration consistency signal. A small PyTorch autoencoder then reconstructs customer-history features and chemical consistency features; a large reconstruction error produces an anomaly flag and the three largest per-feature errors. Transaction anomaly labels are excluded from autoencoder fitting, stopping, architecture selection and threshold calibration.

Chemical profiling foundation

Milestone 1 remains the supervised three-class wine cultivar task based on 178 UCI Wine samples and 13 chemical measurements [1]. Its source CSV and seed-42 sample assignments are preserved. The existing trained chemical network is retained, rather than replaced by the autoencoder. Out-of-fold chemical predictions supply the transaction training features. The new transaction layer uses simulated customer event histories; it does not reinterpret chemical IQR flags as confirmed fraud.

Current experiment

The generator creates 30,000 timestamped events for 1,000 synthetic customers. The first five events per customer establish history, leaving 25,000 scored purchases. Behavioural features are calculated using only earlier events. Both customer identities and linked wine samples are disjoint across train, validation and test. A transaction-only AE is the chemistry ablation. Transaction-only MLP, combined linear and weighted/unweighted combined MLP comparisons use the same features and partitions.

Held-out test result	Primary unsupervised AE	Weighted MLP
Scored purchases / anomalies	3,750 / 280	Same test records
Anomaly precision	0.474	0.944
Anomaly recall	0.329	0.668
False alarms / missed anomalies	102 / 188	11 / 93
Macro F1	0.673	0.884

The AE is primary because this experiment studies screening without transaction labels, not because it outperforms the labelled comparator. Its low recall and scenario-specific failures are reported directly. Synthetic labels support evaluation and the supervised comparison; they do not establish real fraud. The previous experiment used different data and its scores are not directly comparable.

Raw events and past-only behavioural features

Simulator and anomaly injection

Each customer has 30 events, a usual price/quantity pattern, a home location and three usual merchant categories. The base price is lognormal around 25 simulation units; inter-event gaps are exponential with a customer mean between 3 and 20 hours. Quantity is Poisson around a customer mean of 2 to 5, with a minimum of one. Timestamps are increasing within each customer. Amount is unit price multiplied by quantity. Customer, event and sample IDs are linkage fields, never learned predictors.

Scenario	Change to raw event stream
Amount	Multiply unit price by 0.12 or 7.5 and add 12-29 units.
Velocity	Create short 3-15 second gaps and subsequent burst events. The recent count is derived later.
Merchant	Use a category not previously observed for that customer.
Location	Move latitude and longitude by several degrees; derive distance from the previous event.
Chemical mismatch	Change the declared class away from the source cultivar with probability 0.97.
Combined	Apply amount, velocity, merchant, location and declaration changes together.

The nominal scenario-start probability is 0.02 for training and 0.05 for validation/test. Burst continuation increases the observed anomaly rates. Scenario selection probabilities are amount 0.20, velocity 0.20, merchant 0.15, location 0.15, chemical mismatch 0.20 and combined 0.10. The first five events are normal warm-up events. Rare legitimate novelty/travel and incidental declaration errors each occur with probability 0.003. Returning from a synthetic remote location can also look unusual without being labelled anomalous.

Nine numeric model inputs

Feature	Definition at the current event
Amount deviation	Current log(amount) minus prior mean, divided by prior standard deviation, floor 0.25; clip to +/-15.
Quantity deviation	Same construction with log(1 + quantity), scale floor 0.20; clip to +/-15.
Velocity / elapsed time	log(1 + prior events in the last hour); log(1 + minutes since previous event).
New merchant / location shift	Category unseen in prior history; log(1 + haversine distance from previous location in km).
Unit price	Log of current unit price.
Chemical consistency (two)	1 - P(declared cultivar | chemistry); predicted cultivar != declared cultivar.

Every history changes after deriving the current features. No future event or scenario label enters a feature. All customers receive the same five-event warm-up rule, independent of anomaly labels. Python source records the complete reproducible generator and state-update procedure.

Partitions, contamination and leakage control

Partition	Customers	Raw events	Scored events	Anomalies
train	700	21000	17500	553 (3.16%)
validation	150	4500	3750	279 (7.44%)
test	150	4500	3750	280 (7.47%)

Customer and chemical sample separation

The original stratified chemical split is retained: 124 training, 27 validation and 27 test wine samples. Seven hundred training customers draw only from the training wines, and each set of 150 held-out customers draws only from its assigned held-out wines. Thus no customer identity or wine sample crosses partitions. Extra events do not create additional independent chemical samples. Random customer holdout evaluates new simulated customers; it is not a temporal or real-world external holdout.

Strictly past-only feature calculation

Events are sorted within each customer. Prior means, variances, categories, last location and last-hour event counts are read before the current event updates state. Histories begin inside each partition and need no information from another partition. A test customer has its own warm-up period. Feature scaling is fitted only on the 17,500 scored training transactions, including their unlabelled contamination; validation/test use the frozen mean and standard deviation.

Chemical model and out-of-fold predictions

The retained 13-16-3 chemical model is the previous verified checkpoint, with its training-only preprocessing. The chemical task remains supervised. To avoid in-sample chemical predictions when fitting the transaction models, training wines receive three-fold out-of-fold predictions. Each fold model uses a separate internal stratified stopping subset. The final retained chemical model supplies validation and test probabilities. Chemical validation labels were used in the earlier chemical development; transaction anomaly labels are a different target and never enter the AE.

Unlabelled training with contamination

The AE trains on all 17,500 scored training records, including 553 simulated anomalies (3.16%). It is not trained on a normal-only subset selected with labels. The generator knows its scenario labels, but AE fitting and selection receive only numeric features. The validation set is also contaminated (7.44%). Consequently the AE learns the dominant training structure; it cannot be assumed to learn an uncontaminated normal distribution or to assign large errors to every anomaly.

Frozen evaluation and threshold

The bottleneck is selected from widths 2 and 3 using minimum validation reconstruction loss. The primary decision threshold is the predeclared 95th percentile of mixed validation reconstruction scores: 4.424379. This targets approximately 5% validation review volume, not 5% false positives and not a guaranteed test alert rate. Anomaly labels are used only after the AE choices are frozen for metrics and per-scenario analysis. Supervised MLP comparisons may use training/validation labels but never test labels for selection.

Model families, architectures and configuration

Family	Fit and role	Limitations
PyTorch autoencoder	Primary unsupervised detector; compact numeric vectors and reconstruction explanations.	Can reconstruct anomalies, especially with contamination; feature scaling and bottleneck matter.
Linear classifier	Fitted 9-input supervised baseline; small, simple decision surface and cheap inference.	Cannot represent nonlinear interactions without explicit feature engineering.
Supervised MLP	Retrained labelled comparator for the same synthetic scenarios.	Requires anomaly labels and can learn generator-specific patterns.

Selected architecture and inductive bias

The primary unsupervised detector is 9 inputs -> 8 ReLU -> 3 ReLU bottleneck -> 8 ReLU -> 9 linear outputs, with 220 parameters. It minimizes the mean squared standardized reconstruction error [2]. The bottleneck encourages shared structure among purchase behaviour and chemical consistency features; there is no spatial or sequence architecture assumption. History is represented by past-only derived inputs rather than a recurrent network. The transaction-only ablation is 7-8-3-8-7 (186 parameters).

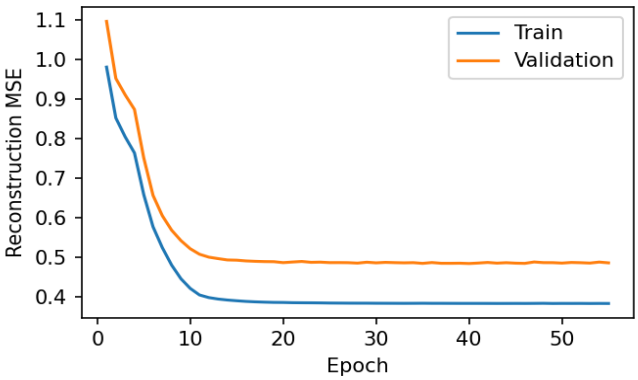
Anomaly score s(x) = mean over features of (z_j - reconstructed z_j)^2, where z is the training-standardized vector. The score is unbounded and is not a fraud probability. The combined supervised MLP is 9-16-2 with ReLU, integer targets and cross-entropy (194 parameters), with weighted and unweighted versions. The transaction-only MLP is 7-16-2 (162 parameters); the combined linear classifier is 9-2 (20 parameters). Their thresholds are selected on validation macro F1 from 37 values between 0.05 and 0.95.

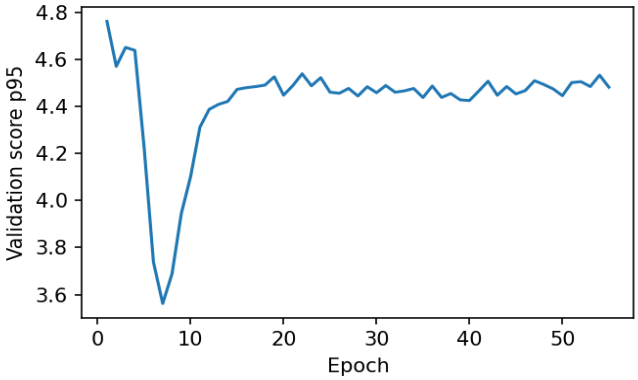
Versioned setting	Value
Configuration / seed	3.1-complete-comparisons / 42
Optimizer / learning rate	Adam / 0.001
Weight decay / batch size	0.0001 / 256
Epoch cap / early stopping	150 / patience 15; minimum loss improvement 0.00001; restore best state
AE selection grid	Bottleneck 2 or 3; validation reconstruction MSE only
AE loss / threshold	MSE / mixed-validation 95th percentile
MLP imbalance mitigation	Training inverse-frequency weights; unweighted comparison
Augmentation / resampling	None; no labels used to remove AE training records

Bottleneck-2 minimum validation MSE was 0.506012, versus 0.484416 for bottleneck 3. These development losses select capacity, not anomaly detection accuracy. The lowest reconstruction loss need not yield the best anomaly separation; selecting by anomaly F1 would introduce labels into this unsupervised selection. The chosen small network limits parameter memory and dense arithmetic. Its expected label-free benefit is evaluated against the measured supervised comparison, rather than assumed.

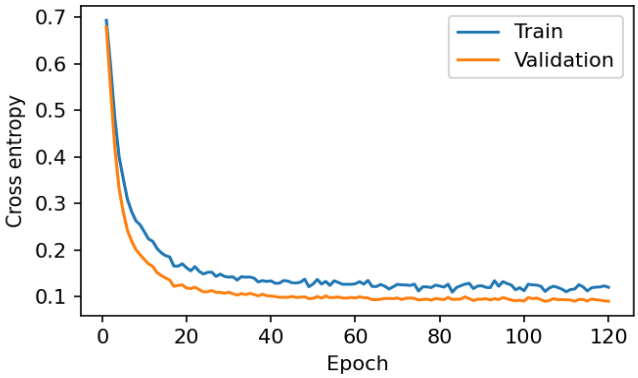
Training execution and learning curves

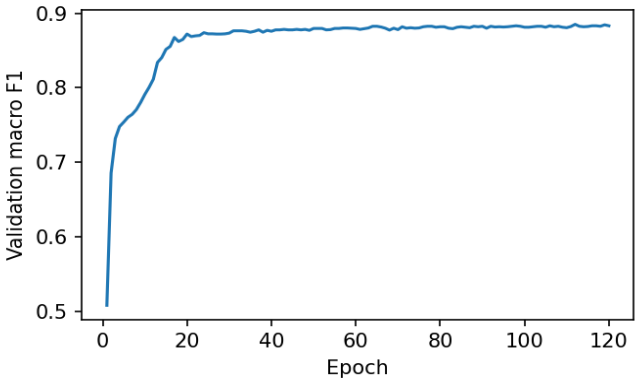
Primary AE: training and validation reconstruction





Weighted supervised comparison





Model	Executed / best epoch	Seconds	Seconds / epoch
ae_combined_b2	80 / 65	3.590	0.045
AE + chemistry	55 / 40	2.571	0.047
AE: transactions only	84 / 69	3.681	0.044
Weighted MLP	120 / 105	3.925	0.033
Unweighted MLP	150 / 140	4.814	0.032
Transaction-only MLP	64 / 49	1.806	0.028
Combined linear	86 / 71	1.822	0.021

AE curves show train/validation mean reconstruction loss and the 95th percentile of validation scores. The MLP metric curve is validation macro F1 at argmax, while final decisions use its separately selected validation threshold. Patience epochs remain visible; the best checkpoint is restored. Times include epoch validation and history calculations. The chemical final checkpoint is reused; OOF chemical fits are included in the data-preparation time, not in the transaction model rows.

Hardware: AMD EPYC 9V74 80-Core Processor, one PyTorch compute thread, 9 reported logical CPUs, 9.73 GiB RAM, no GPU. Python 3.12.14; PyTorch 2.14.1+cpu. All CSV epoch histories and comparison-model curves are included.

Validation and held-out test performance

Primary AE metric	Validation	Test
Accuracy	0.924	0.923
Anomaly precision	0.484	0.474
Anomaly recall	0.326	0.329
Anomaly F1	0.390	0.388
Macro precision	0.716	0.711
Macro recall / balanced accuracy	0.649	0.650
Macro F1	0.675	0.673
Micro F1	0.924	0.923
AUROC	0.841	0.835
PR-AUC (average precision)	0.454	0.441

Same-data comparisons and imbalance effect

Model	Macro F1	Anomaly F1	PR-AUC	FP / FN
AE + chemistry	0.673	0.388	0.441	102 / 188
AE: transactions only	0.655	0.353	0.459	112 / 196
Weighted MLP	0.884	0.782	0.907	11 / 93
Unweighted MLP	0.876	0.769	0.905	24 / 90
Transaction-only MLP	0.884	0.783	0.823	6 / 96
Combined linear	0.812	0.658	0.700	173 / 58

The primary AE confusion matrix is [[3368, 102], [188, 92]], with rows actual normal/anomalous and columns predicted normal/anomalous. It flags 194/3,750 purchases (5.17%), of which 102 are false alarms. The test set contains 280 anomalies; 188 are missed. Always predicting normal would give 92.53% accuracy but zero anomaly recall. AUROC and average precision use continuous scores, independently of the fixed threshold [3].

The combined AE modestly improves thresholded macro F1 over the transaction-only AE, but has lower AUROC and PR-AUC. Chemistry is therefore not shown to improve every detection measure. The weighted supervised MLP performs better than the AE on this labelled simulator. Weighting raises precision from 0.888 to 0.944 and reduces false alarms from 24 to 11 relative to the unweighted MLP, while recall falls from 0.679 to 0.668 after separate validation threshold selection.

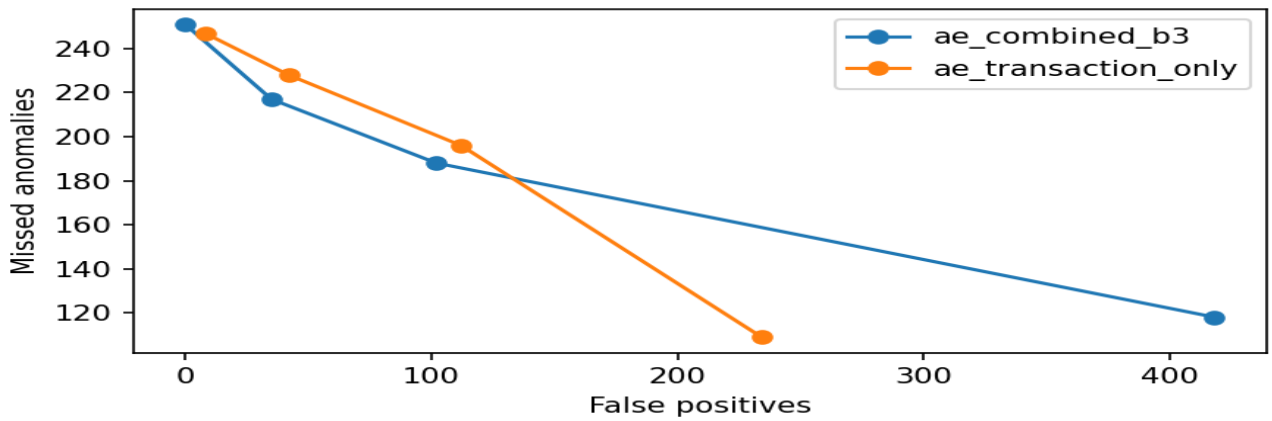
The AE remains the primary research design because its transaction fitting requires no anomaly labels. This choice does not establish operational superiority. The 3.16% training contamination, correlated feature groups and the difference between training and held-out scenario prevalence limit reconstruction-based discrimination. Test results are reported once after choices are frozen and are not used to alter the model.

Threshold trade-offs and scenario failures

Validation percentile	Threshold	Test precision	Test recall	FP / FN
90.0	0.429	0.279	0.579	418 / 118
95.0	4.424	0.474	0.329	102 / 188
97.5	4.958	0.643	0.225	35 / 217
99.0	6.215	1.000	0.104	0 / 251

All threshold values are obtained from validation scores; the table describes their test consequences without selecting a new operating point. The 95th-percentile operating point stays fixed. A more permissive threshold increases detection and review burden; a stricter threshold misses more anomalies. The percentile is a queue-capacity policy, not an estimate of class prevalence.

Test scenario	Count	AE + chemistry recall	AE transaction recall	Weighted MLP recall
amount	28	0.571	0.000	1.000
velocity	132	0.023	0.053	1.000
merchant	30	0.933	1.000	0.067
location	28	0.929	1.000	0.214
chemical_mismatch	43	0.000	0.000	0.000
combined	19	1.000	1.000	1.000



At the chosen threshold the AE catches most merchant and location cases, but only 3/132 velocity cases and no chemical-mismatch-only cases. The network can reconstruct chemical inconsistencies and some burst structures, so these signals do not guarantee high scores. Combined scenarios are all flagged. The supervised model catches all amount and velocity cases, but misses many merchant, location and chemical mismatch cases at its selected threshold. The per-scenario results quantify these failure modes. results/per_scenario_recall.csv provides the complete six-model scenario comparison.

Reconstruction explanations for flagged records

For standardized input z and reconstruction z-hat, per-feature error is (z_j - z-hat_j)^2. The anomaly score is the mean of the nine errors. The three largest errors are reported for each flagged transaction. Error shares divide each squared error by the sum, so they describe the fraction of reconstruction error, not a probability of guilt. All 194 flagged test records have saved explanations.

Example 1: novel merchant category

Transaction test-C0005-T08: score 5.181547, threshold 4.424379; decision: human review. Generator scenario: merchant, shown only for evaluation.

Feature	Squared standardized error	Share
new merchant category	42.016	90.1%
log unit price	3.637	7.8%
log location shift km	0.495	1.1%

Example 2: location shift

Transaction test-C0010-T06: score 5.283775, threshold 4.424379; decision: human review. Generator scenario: location, shown only for evaluation.

Feature	Squared standardized error	Share
log location shift km	45.694	96.1%
log minutes since previous	1.261	2.7%
log unit price	0.279	0.6%

The first record is difficult to reconstruct mainly because its merchant category is new to that customer. The second is dominated by distance from the previous location. These statements describe the model output, not verified misconduct. Legitimate travel, a new merchant or contamination can create the same signal. Source features and complete event prefixes are included for replay.

Explanation boundaries

Squared reconstruction error gives a direct score decomposition, but correlated features may double-count a behavioural change and the training scaler determines relative units. Large reconstruction error does not reveal a causal mechanism. No SHAP or causal attribution is claimed. The examples complement the complete per-scenario analysis on page 7. The notebook contains model scores, the scenario comparison and a raw-history replay.

Live demonstration input and output

demo.py accepts a synthetic customer and an optional event index. It rebuilds state from that customer's earlier raw records, derives current features, runs chemical inference and the AE, and returns score, threshold, decision and the three largest errors. The first five events return history_warmup. Invalid sample/classes, non-finite inputs, inconsistent purchase amounts and repeated/out-of-order timestamps are rejected. No test labels are needed by this serving path.

Measured training and inference efficiency

Model	Parameters	Dense forward FLOPs	Training GFLOPs estimate
ae_combined_b2	203	352	1.478
AE + chemistry	220	384	1.109
AE: transactions only	186	320	1.411
Weighted MLP	194	352	2.218
Unweighted MLP	194	352	2.772
Transaction-only MLP	162	288	0.968
Combined linear	20	36	0.163

Forward FLOPs count two operations per dense multiply-accumulate and exclude biases, nonlinearities, scaling and history calculation. Training compute estimates multiply forward FLOPs by three, by training records and by executed epochs (forward plus approximately two forward-equivalents for backward); they exclude optimizer, validation and chemical OOF work. These are analytical counts, not hardware throughput measurements. The primary AE is 384 FLOPs per scored vector (0.000000384 GFLOPs). Including the retained chemical network adds 512 dense FLOPs, for 896 total before non-dense operations.

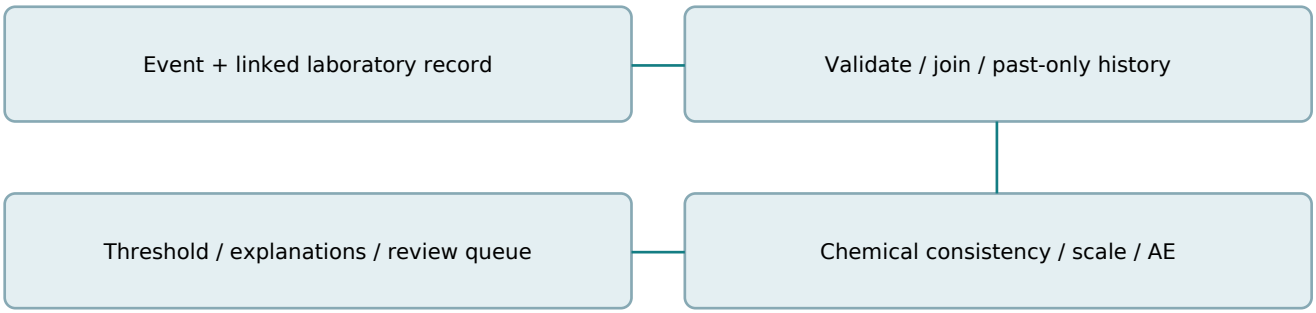
The initial experiment took 25.16 seconds; the added baseline run took 5.01 seconds. The initial run includes 5.63 seconds for chemical OOF fitting and event/feature generation. Summed transaction fitting time is 22.21 seconds. Initial-run peak process RAM sampled every 5 ms was 487.7 MiB. The shared detector checkpoint is 14,033 bytes and includes six model states and preprocessing; the retained chemistry checkpoint is 4,401 bytes. No GPU/VRAM, energy or cost measurement was used.

CPU model / batch	p50 ms	p90 ms	p95 ms	Samples/sec
AE + chemistry / 1	0.038	0.059	0.098	21,528
AE + chemistry / 32	0.045	0.078	0.122	567,668
AE: transactions only / 1	0.037	0.044	0.063	24,018
AE: transactions only / 32	0.043	0.063	0.082	613,931
Weighted MLP / 1	0.020	0.020	0.027	45,483
Weighted MLP / 32	0.025	0.026	0.041	1,167,380

Fresh inference-process RAM after imports was 270.5 MiB; after all benchmark models, data and the raw-event demo loaded it was 350.5 MiB. The primary scoring stage used 285.5 MiB process-wide. These values include runtime and data, not isolated weights. Model timings include standardization and scoring on ready features; batch-32 latency is total batch latency, not per sample.

The raw-event path including history updates, chemical inference, AE scoring and explanations measured p50/p90/p95 0.190/0.282/0.340 ms, about 4,673 events/sec. Both benchmarks use 100 warm-ups and 1,000 timed calls on CPU. File I/O, network and laboratory acquisition are excluded. Quantization, distillation and ONNX export were not attempted; no effects are claimed.

Deployment design and system trade-offs



Online/offline; hybrid edge/cloud; batch/streaming

Offline jobs generate data, fit chemical preprocessing/OOF predictions, train transaction models, calibrate thresholds and validate versioned bundles. Online scoring reads the event and current customer state, derives features before updating state, joins chemical consistency and returns score/review reasons. A laboratory workstation can cache chemical probabilities by sample and measurement/model version; a central service maintains cross-device customer history and the review queue. Connectivity and privacy favour keeping raw laboratory records local; central event history enables velocity detection. This is a proposed hybrid architecture, not a deployed cloud service.

Sequential customer events use batch 1 with per-customer ordering preserved. Cross-customer microbatches up to 32 can improve throughput without mixing history states. Batch reconciliation is offline. Review processing is asynchronous, and cached chemical outputs are invalidated after record or model changes. Event IDs support idempotency. Late/out-of-order events require replay or manual review, never silent history mutation. The local demo implements state and input checks; shared storage, queues and API hosting are future components.

Failure handling and rollback

An unavailable laboratory sample or insufficient customer history produces a pending/manual-review response. A transaction-only AE may be an explicitly marked fallback using its own frozen threshold; coverage differs. The proposed service retains the last validated model/scaler/threshold bundle for atomic rollback after schema mismatches, score spikes or latency failures. Failed jobs go to a bounded retry/dead-letter queue in the proposed service. No automatic fraud conviction or product-safety decision follows from a flag.

Proposed SLO / gate	Evidence or current limitation
Warm raw-event p95 <= 10 ms	Observed 0.340 ms; excludes network and lab acquisition.
API p95 <= 100 ms	Design target; hosted API/load test not implemented.
Anomaly recall >= 0.80	Not met: primary AE recall 0.329 on synthetic holdout.
False-positive rate <= 0.03	Observed 0.029 at the fixed threshold; external validation still needed.

A shadow pilot would log scores without blocking purchases. Monitoring covers p95 latency, schema failures, missing links, history freshness, score/feature drift, flag rate and queue backlog. Reviewed outcomes support precision, recall and false-positive analysis by scenario and cohort. The threshold controls review load; lowering it increases missed-case recovery and false alarms. Future threshold calibration requires separate development outcomes; the reported test labels remain excluded from selection.

Reproducibility and verification

Execution environment and artifacts

The experiment was executed with Python 3.12 and the dependency versions recorded in requirements.txt and environment_versions.txt. The experiment, efficiency, verification and report-generation scripts support reproduction of the study. The command-line and notebook demonstrations load saved checkpoints. Data, model settings, predictions and learning histories accompany the report.

Verification confirmed the source checksum and original sample assignments, disjoint customer and sample groups, past-only feature calculation, label-independent inputs, training-only scaling, reproducible scores and thresholds, raw-event replay, rejection of invalid event ordering, and complete explanation coverage. All six models were evaluated on the same held-out records. Each saved prediction corresponds to its recorded checkpoint and preprocessing parameters.

Artifact	Purpose
config.json; environment_versions.txt	Executed configuration record and package versions; training settings are defined in the scripts.
data/events_*.csv; data/features_*.csv	Raw customer histories and their derived numeric inputs.
models/chemistry.pt; models/detectors.pt	Chemical model and six transaction model states, with preprocessing parameters and thresholds.
results/metrics.json; efficiency.json	Validation/test results, training records and inference measurements.
Prediction and learning-history CSVs	Per-record scores/decisions and epoch-level training evidence.
flag_explanations.json; verification.txt	Reconstruction factors for every flagged test record and verification results.

Conclusion

On this labelled simulator, supervised classification is the stronger operational choice. The combined weighted MLP achieves average precision 0.907 and macro F1 0.884; the transaction-only MLP has the highest macro F1 (0.884235 versus 0.883808). The autoencoder remains the primary unsupervised detector for label-free screening, with complementary merchant and location sensitivity. Chemistry is an auxiliary signal with mixed benefits. The AE misses 188 anomalies and does not meet the proposed recall gate, so deployment requires further validation and improvement.

References

[1] Aeberhard and Forina. Wine [Dataset]. UCI Machine Learning Repository, DOI 10.24432/C5PC7J; CC BY 4.0. https://archive.ics.uci.edu/dataset/109/wine

[2] PyTorch. MSELoss documentation. https://docs.pytorch.org/docs/stable/generated/torch.nn.MSELoss.html

[3] scikit-learn. average_precision_score documentation. https://scikit-learn.org/stable/modules/generated/sklearn.metrics.average_precision_score.html